ANALYZING AND PREDICTING URBAN DEVELOPMENT AND URBAN HEAT ISLAND (UHI) PATTERNS IN ENUGU METROPOLIS (2000 - 2045) USING DEEP MACHINE LEARNING

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UNIVERSITY OF NIGERIA, ENUGU CAMPUS

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TITLE PAGE

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A DISSERTATION SUBMITTED TO THE DEPARTMENT OF GEOINFORMATICS AND SURVEYING, FACULTY OF ENVIRONMENTAL STUDIES, UNIVERSITY OF NIGERIA, ENUGU CAMPUS, IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF MASTER OF SCIENCE (M.Sc.) DEGREE IN GEOINFORMATICS AND SURVEYING

SUPERVISOR: PROFESSOR VICTOR C. NNAM, FNIS

OCTOBER, 2026

CERTIFICATION

This is to certify that OCHOLI ISRAEL (Reg. No: **PG/M.Sc/23/97632**), a postgraduate student in the Department of Geoinformatics and Surveying, Faculty of Environmental Studies, University of Nigeria, Enugu Campus, has satisfactorily completed the requirements for research work for the award of the Degree of Master of Science (M.Sc.) in Geoinformatics and Surveying. The work embodied in this dissertation is original and has not been submitted in part or in full for any other diploma or degree of this or any other University.

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DEDICATION

This dissertation is dedicated to Almighty God for His infinite grace, wisdom, and strength throughout this academic journey, and to my beloved family for their unwavering support, prayers, and encouragement.

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My profound gratitude goes first to Almighty God for granting me the health, perseverance, and intellect to successfully complete this Master of Science programme.

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ABSTRACT

Rapid and largely unplanned urbanization in Sub-Saharan African secondary cities has accelerated the conversion of natural vegetative landscapes into impervious surfaces, intensifying the Urban Heat Island (UHI) phenomenon. This study applied an integrated Deep Machine Learning, Remote Sensing, and Geographic Information Systems (GIS) framework to analyze historical Land Use/Land Cover (LULC) transformations and UHI dynamics in Enugu Metropolis from 2000 to 2024, and to forecast future spatial and thermal trajectories up to the year 2045. Multi-temporal satellite imagery from Landsat 5 (TM), Landsat 7 (ETM+), Landsat 8 (OLI/TIRS), and Sentinel-2 (MSI) for the epochs 2000, 2005, 2010, 2015, 2020, and 2024 were radiometrically calibrated, cloud-masked, and harmonized at a 30-meter spatial resolution within Google Earth Engine (GEE). Five spectral indices (NDVI, NDBI, BSI, MNDWI, and NDMI), emissivity-corrected Land Surface Temperature (LST), topographic slope (SRTM DEM), and Euclidean road proximity (OpenStreetMap) were stacked into 16-band tensors and tiled into 32×32 pixel spatial chips. To resolve the multi-task gradient conflict between discrete physical land-cover boundaries and continuous thermal diffusion, a customized Hybrid Dual-Head Convolutional Long Short-Term Memory (ConvLSTM) architecture was developed in TensorFlow/Keras: Head A performed categorical LULC segmentation using a Softmax activation and cross-entropy loss, while Head B executed continuous thermodynamic regression (Mean Squared Error loss) prior to standardized Z-score thresholding (±1.5). Empirical findings revealed that Total Built-Up land nearly doubled over the 24-year historical period, expanding by +77.27% (+36.65 km²) from 47.43 km² in 2000 to 84.08 km² in 2024 at an annual expansion rate of +1.53 km²/year, driven primarily by highway corridor accessibility (81.4% of new growth within ≤1.5 km of major arterial roads) and constrained westward by the steep Milliken Hill escarpment (92.6% of growth on slopes <6°). Concurrently, metropolitan Mean LST rose by +3.14°C, from 35.60°C in 2000 to 38.74°C in 2024. Model validation against 2024 ground-truth reference samples demonstrated that the Hybrid ConvLSTM achieved an Overall Accuracy of 87.12% ($\kappa = 0.701$) for LULC classification (and 87.12%–89.03% across historical epochs) and an R^2 of 0.7734 (72.64% post-thresholding accuracy, $\kappa = 0.550$) for UHI prediction, substantially outperforming Maximum Likelihood Classification (67.39% OA), Support Vector Machines (75.80% OA), Random Forest (78.45% OA), and CA-Markov spatial allocation (K-Location = 0.8815 vs. 0.6890). Spatio-temporal forecasts to 2045 project that under business-as-usual conditions, Total Built-Up area will expand by an additional +69.60% (+58.52 km²) to reach 142.60 km² at an accelerated rate of +2.79 km²/year, driving Mean LST to 41.35°C (+2.61°C) and more than tripling the severe UHI Hotspot footprint (+203.61%) from 22.99 km² in 2024 to 69.80 km² by 2045. The study recommends urgent revision of the Enugu Master Plan, enforcement of urban green belts along riparian and escarpment buffers, and institutional adoption of AI-driven spatial decision support systems.

TABLE OF CONTENTS

Title Page i

Certification ii

Dedication iii

Acknowledgements iv

Abstract v

Table of Contents vi

List of Tables xi

List of Figures, Maps, Charts, and Curves Xiii

List of Abbreviations and Acronyms xiv

CHAPTER ONE: INTRODUCTION 1

1.1 Background to the Study ----------1

1.2 Statement of Problem ----------4

1.3 Research Objectives ----------5

1.4 Research Questions ----------6

1.5 Significance of the Study ----------6

1.6 Study Area ----------7

CHAPTER TWO: LITERATURE REVIEW 10

Theoretical Framework ----------10

2.1 Urban Development Patterns Theories ----------10

2.2 Urban Heat Island (UHI) ----------14

2.2 (2.2.1) Urban Ecology Theory ----------16

2.3 GIS in Urban Planning ----------17

2.4 Remote Sensing of Land Surface Temperature (LST) ----------19

2.5 Deep Machine Learning and Spatial Prediction Models ----------21

2.6 Urbanization in Nigeria and Enugu ----------22

2.7 Predictive Modeling Techniques ----------25

2.8 Empirical Review ----------27

2.8.1 Global Applications of Deep Learning in Urban Analysis ----------30

2.9 Research Gap ----------32

CHAPTER THREE: RESEARCH METHODOLOGY **33**

3.1 Research Design ----------34

3.2 Data Collection and Data Source ----------36

3.3 Data Processing and Feature Engineering ----------37

3.3.1 Radiometric Calibration and Cloud Masking ----------37

3.3.2 Multi-Sensor Harmonization (2015–2024) ----------38

3.3.3 Derivation of Spectral Indices and LST (Equations 3.1–3.11) ----------38

3.3.4 Image Tiling and Tensor Extraction ----------40

3.3.5 Target Variable Formatting (Continuous vs. Categorical Alignment) ----------41

3.4 Data Analysis ----------42

3.4.1 LULC Spatio-Temporal Change Detection (Equations 3.12–3.15) ----------43

3.4.2 Thermodynamic Analysis of Urban Heat Islands (UHI) (Equations 3.16–3.17) ----------44

3.4.3 Predictive Modeling and Forecasting to 2045 ----------45

3.4.4 Statistical Validation and Model Evaluation ----------45

3.5 Deep Learning Model Design ----------46

3.5.1 Input Pipeline and Shared Feature Extraction ----------46

3.5.2 The Dual-Head Architecture (Equations 3.18–3.22) ----------47

3.5.3 Model Training and Optimization ----------48

3.5.4 Model for Extracting Urban Development Information (U-Net-CNN) ----------49

3.5.5 Model Training, Validation, and Testing ----------50

3.6 Land Use Classification & UHI Analysis ----------50

3.6.1 Land Use and Land Cover (LULC) Change Detection (2002–2024) ----------50

3.6.2 Urban Heat Island Zoning and Spatial Statistics ----------50

3.7 Model for Prediction and Comparing Changes in Urban Development and UHI ----------51

3.7.1 Spatio-Temporal Forecasting using ConvLSTM ----------51

3.7.2 Integration of UHI Thermodynamic Prediction ----------52

3.7.3 Comparative Evaluation of Growth Models (Equation 3.23) ----------53

3.7.4 Quantifying Projected Changes ----------53

3.8 Statistical Evaluation of Model Performance (Equations 3.24–3.30) ----------54

CHAPTER FOUR: RESULTS AND DISCUSSION **56**

4.1 Analysis of Urban Development and UHI Patterns (2000–2024) ----------56

4.1.1 Historical LULC Area Distribution ----------56

4.1.2 Historical Land Surface Temperature (LST) and UHI Dynamics ----------60

4.1.3 Key Topographic, Infrastructural, and Socioeconomic Drivers of Urban Expansion ----------66

4.2 Application and Performance of the CNN-Driven Model ----------68

4.2.1 Overall Performance Metrics and Training History ----------69

4.2.2 Model Validation Metrics (2024) ----------72

4.3 Prediction of Future Urban Expansion and UHI Hotspots up to 2045 ----------85

4.3.1 Quantitative Analysis of Transition Probabilities and Projected LULC Expansion ----------85

4.3.2 Projected Urban Heat Island (UHI) Intensification and Hotspot Emergence (2030–2045) ----------88

4.4 Evaluation of Deep Learning vs. Conventional Methods ----------92

4.4.1 Robustness to Sensor Shifts and Traditional Classifiers ----------92

4.4.2 Categorical vs. Continuous Thermodynamic Modeling ----------92

4.4.3 Benchmarking Against Random Forest, SVM, and CA-Markov Spatial Forecasting ----------93

CHAPTER FIVE: DISCUSSION OF FINDINGS, CONCLUSION, AND RECOMMENDATIONS **95**

5.1 Discussion of Findings ----------95

5.1.1 Urban Expansion and Ecological Degradation ----------95

5.1.2 The UHI Phenomenon and Methodological Advancements ----------96

5.2 Conclusion ---------- 98

5.3 Recommendations ----------98

REFERENCES **100**

LIST OF TABLES

Table 1 (Table 3.1): Summary of Application of Models	54
Table 4.1: Historical LULC Area Distribution (2000–2024)	57
Table 4.1b: Spatio-Temporal LULC Change Detection Metrics (ΔA, Pc, and Annual Expansion Rate Re, 2000–2024)	57
Table 4.2: Historical Mean LST and UHI Zone Distribution (2000–2024)	62
Table 4.2b: Quantitative Evaluation of Key Drivers of Urban Expansion and Thermal Intensification in Enugu Metropolis (2000–2024)	66
Table 4.3: Epoch-by-Epoch Training and Validation Metrics	70
Table 4.4: LULC Classification Accuracy Metrics for 2024 Validation Baseline (Head A)	72
Table 4.5: UHI Thermal Zone Classification Accuracy Metrics for 2024 Validation Baseline (Head B)	72
Table 4.6a: Error Matrix of Land Use/Land Cover (LULC) Classes (Head A) for Enugu Metropolis (2000–2024)	73
Table 4.6b: Accuracy Metrics Summary (Land Use/Land Cover (LULC) Classes (Head A)) for Enugu Metropolis (2000–2024)	75
Table 4.7a: Error Matrix of Urban Heat Island (UHI) Thermal Zones (Head B) for Enugu Metropolis (2000–2024)	80

Table 4.7b: Accuracy Metrics Summary (Urban Heat Island (UHI) Thermal Zones (Head B)) for Enugu Metropolis (2000–2024) 81

Table 4.8: Predicted Land Use Transition Probability Matrix for 2045 86

Table 4.8a: Projected LULC Area Distribution and Sprawl Rate Metrics (2024 Baseline – 2045) 86

Table 4.8b: Projected Mean LST and UHI Thermal Zone Distribution (2024 Baseline – 2045) 88

Table 4.9: Comprehensive Comparative Evaluation: Hybrid ConvLSTM vs. Conventional Machine Learning (RF, SVM, MLC) and Spatial Forecasting (CA-Markov) Baselines 93

LIST OF FIGURES, MAPS, CHARTS, AND CURVES

Figure 1: Map of Enugu Showing the Study Area 8

Figure 3: The Flow Chart of the Methodology 33

Figure 3.1: Methodology Workflow (Four-Phase Deep Learning and Cloud Extraction Pipeline) 35

Figure 3.2: Google Earth Engine Scripts 37

Figure 3.3.4: VS Code / Jupyter Notebook Model Building 41

Chart 4.1: LULC Composition & Internal Shifts (2000–2024) 59

Curve 4.1: Trend of Total Built-Up Area Expansion (2000–2024) 60

Figure 4.2: Urban Development and UHI Patterns (2000–2024 Maps) 63–65

Graph 4.1: Hybrid ConvLSTM Training vs. Validation Accuracy 71

Figure 4.2.2: Validation: 2024 Actual vs. Predicted 84

Figure 4.3: Predicted Land Use Land Cover (LULC) and UHI Forecast Maps of Enugu Metropolis (2030–2045) 90–91

LIST OF ABBREVIATIONS AND ACRONYMS

ABM	Agent-Based Modeling
ANN	Artificial Neural Network
BSI	Bare Soil Index
BT	At-Sensor Brightness Temperature
CA-Markov	Cellular Automata–Markov Chain Model
CBD	Central Business District
CNN	Convolutional Neural Network
ConvLSTM	Convolutional Long Short-Term Memory Network
DEM	Digital Elevation Model
DN	Digital Number
ETM+	Enhanced Thematic Mapper Plus (Landsat 7)
FCN	Fully Convolutional Network
GEE	Google Earth Engine
GIS	Geographic Information Systems
IoU	Intersection over Union (Jaccard Index)
LST	Land Surface Temperature
LULC	Land Use / Land Cover
MLC	Maximum Likelihood Classification
MNDWI	Modified Normalized Difference Water Index
MSE	Mean Squared Error
MSI	MultiSpectral Instrument (Sentinel-2)
NDBI	Normalized Difference Built-up Index
NDMI	Normalized Difference Moisture Index

NDVI	Normalized Difference Vegetation Index
NIR	Near-Infrared Band
NPC	National Population Commission
OA	Overall Accuracy
OLI	Operational Land Imager (Landsat 8/9)
PA	Producer's Accuracy
RF	Random Forest
RMSE	Root Mean Square Error
SRTM	Shuttle Radar Topography Mission
SVM	Support Vector Machine
SWIR	Short-Wave Infrared Band
TIR / TIRS	Thermal Infrared Sensor
TM	Thematic Mapper (Landsat 5)
TOA	Top of Atmosphere Reflectance
UA	User's Accuracy
UHI	Urban Heat Island
UNEC / UNN	University of Nigeria, Enugu Campus / Nsukka
USGS	United States Geological Survey